



Started on	Thursday, 17 September 2026, 8:35 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:41 AM
Time taken	5 mins 7 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 int main() {
3     int n, max = 0;
4     if (scanf("%d", &n) != 1) return 0;
5     int seq[n], dp[n];
6     for (int i = 0; i < n; dp[i] = 1, i++) scanf("%d", &seq[i]);
7     for (int i = 0; i < n; max = (dp[i] > max) ? dp[i] : max, i++)
8         for (int j = 0; j < i; j++)
9             if (seq[j] <= seq[i] && dp[i] < dp[j] + 1) dp[i] = dp[j] + 1;
10    printf("%d\n", max);
11 }
12

```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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